ODIR-5K Multimodal Disease Classification

End-to-end machine learning project report | Fundus images + age + sex | Eight classes

1. Executive summary

This project developed and compared multimodal deep-learning models for classifying ocular conditions from ODIR-5K retinal fundus images and clinical features (standardized age and sex indicators). The selected EfficientNetB0 model achieved **52.48% test accuracy**, **0.4369 macro F1**, and **0.507 weighted F1** on 966 held-out images. Performance varied substantially across classes; this is an educational research prototype, not a clinical diagnostic system.

2. Dataset and preparation

The working dataset contained 6,392 images. Class counts: N/Normal 2,873 (44.95%); D/Diabetes 1,608 (25.16%); O/Other abnormalities 708 (11.08%); C/Cataract 293 (4.58%); G/Glaucoma 284 (4.44%); A/AMD 266 (4.16%); M/Myopia 232 (3.63%); H/Hypertension 128 (2.00%). Images were resized to 224 × 224 RGB. Age was standardized using training data, and sex was encoded into two indicators.

Patient-level splitting was used to prevent images from the same patient appearing across different splits: training 4,478 images, validation 948 images, and test 966 images. Pairwise patient overlap was zero. Validation results informed model selection; the test set was evaluated after selection.

3. Model architecture and experiments

Each model fused pretrained CNN image features with clinical features. The fusion head used dense layers and dropout and produced eight softmax probabilities. Experiments included frozen MobileNetV2 with full class weights, frozen MobileNetV2 without class weights, fine-tuned MobileNetV2, frozen MobileNetV2 with smoothed class weights, MobileNetV2 with smoothed weights and augmentation, and frozen EfficientNetB0 with smoothed class weights.

Experimen	Configuration	Validation accuracy	Macro F1
1	MobileNetV2 frozen; full weights	41.03%	0.404
2	MobileNetV2 frozen; unweighted	54.64%	0.401
3	MobileNetV2 fine-tuned; unweighted	52.53% restored*	0.385
4	MobileNetV2 frozen; smoothed weights	50.21%	0.451
5	MobileNetV2 + smoothed weights + augmentation	46.73% peak**	Not recorded
6	EfficientNetB0 frozen; smoothed weights	53.16%	0.458

*Fine-tuned MobileNetV2 metrics were from its restored checkpoint. **The augmentation run restored epoch 9 by validation loss; its best observed validation accuracy was 46.73% at epoch 7. EfficientNetB0 was selected based on validation loss and the highest recorded validation macro F1, not accuracy alone.

4. Final held-out test results

Accuracy: 52.48% **Macro F1:** 0.4369 **Weighted F1:** 0.507 **Samples:** 966

Class	Precision	Recall	F1	Support
N — Normal	0.644	0.623	0.633	448
D — Diabetes	0.372	0.519	0.434	214
G — Glaucoma	0.338	0.450	0.386	60
C — Cataract	0.630	0.773	0.694	44
A — AMD	0.522	0.267	0.353	45
H — Hypertension	0.250	0.042	0.071	24
M — Myopia	0.673	0.946	0.787	37

Class	Precision	Recall	F1	Support
O — Other abnormalities	0.364	0.085	0.138	94

5. Interpretation

Myopia recall was 94.6% and cataract recall was 77.3%. Normal recall was 62.3%, diabetes 51.9%, and glaucoma 45.0%. Hypertension recall was 4.2%, and Other-abnormality recall was 8.5%. These results show why accuracy must be read alongside per-class recall, macro F1, and the confusion matrix. The broad Other category may contain heterogeneous conditions, but the experiments do not establish the cause of its low recall.

6. Limitations and responsible use

The model is not suitable for clinical diagnosis, screening, triage, or treatment decisions. Performance is modest and highly uneven across classes. Potential limitations include class imbalance, few examples for rare classes, label quality, image variation, and the broad Other category. Evaluation on ODIR-5K does not demonstrate generalization to other populations or devices. Any clinical use would require external validation, clinical oversight, calibration, and appropriate safety and regulatory review.

7. End-to-end lifecycle completed

Dataset acquisition and path validation; metadata and target preparation; patient-aware train/validation/test split; age/sex preprocessing; image pipeline construction; multimodal model development; experiments with class weighting, fine-tuning and augmentation; validation-based model comparison; final held-out test evaluation; and documentation of limitations.

8. Conclusion

The project completed an end-to-end multimodal classification workflow. The final EfficientNetB0 model achieved 52.48% held-out test accuracy and 0.4369 macro F1. While not clinically adequate, the work demonstrates practical skills in data preparation, leakage-aware splitting, transfer learning, multimodal fusion, imbalanced-class evaluation, model comparison, and responsible reporting. Further optimization against the held-out test set should be avoided.